

Smart Irrigation System



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Problem

Many farmers still use traditional, manual methods to water crops , often relying on guesswork rather than data. This leads to:

- Overwatering or underwatering crops
- Unhealthy plant growth due to poor nutrient distribution

Project Objectives

- Saves water by only irrigating when needed
- Boosts crop health and yield through better moisture and nutrient control
- Reduces manual labor and reliance on guesswork
- Affordable and scalable for both small farms and large plantations

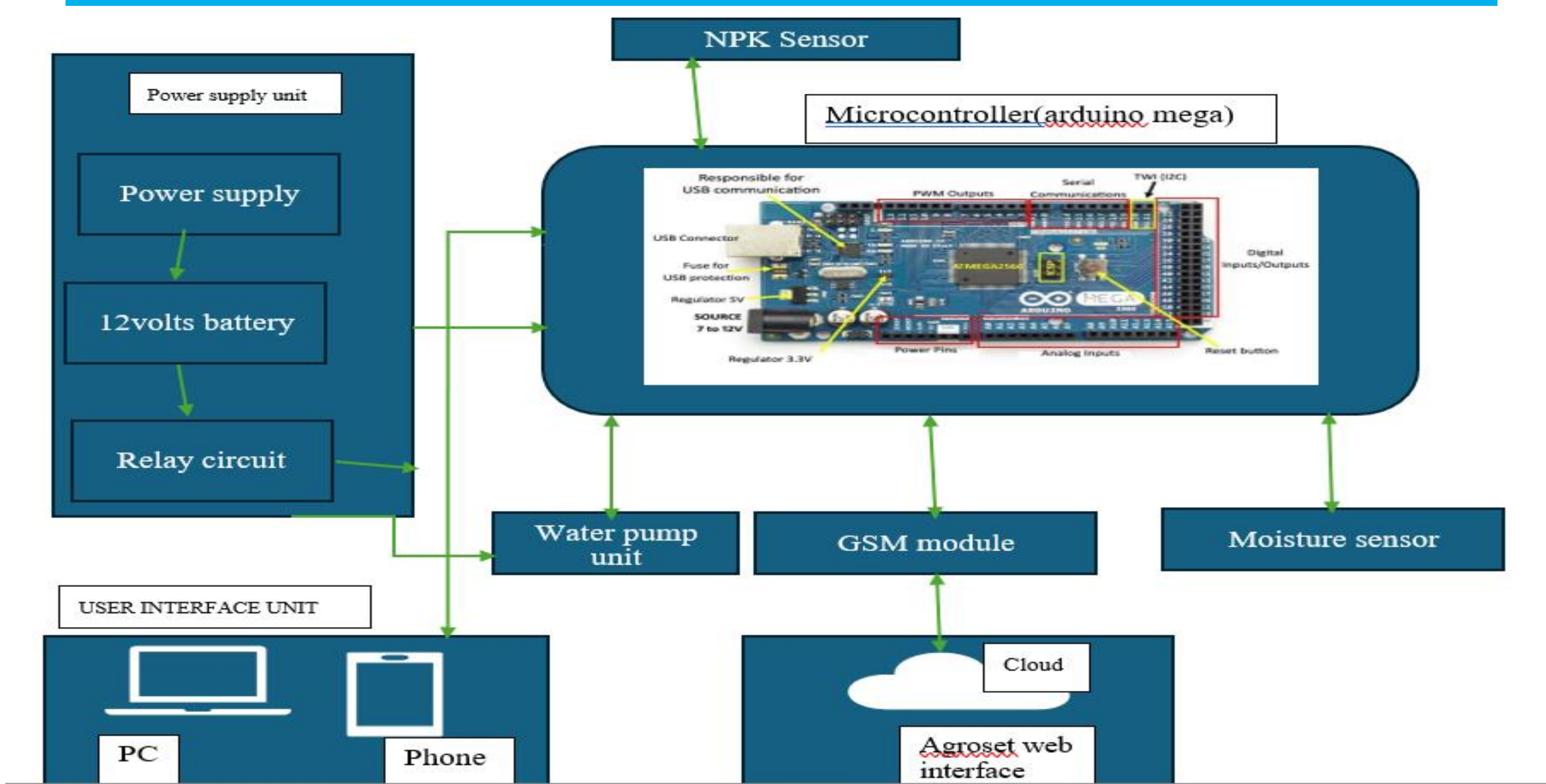
Project Requirements

- Soil moisture sensors to check real-time water levels
- Automatic control of water pumps (ON/OFF) based on soil needs
- Fertilizer injection system for improved crop nutrition
- User-friendly web interface to monitor data and control the system remotely

Target Users

Smallholder Famers	Commercial Farm Owners	Research Institutes &schools
• Easy to use, saves water and waters crops automatically especially in areas with little water or rainfall	• Automates watering and fertilizing, saving costs and increasing farm productivity	• Makes it easy to study how smart watering affects crops and soil health

Project Design

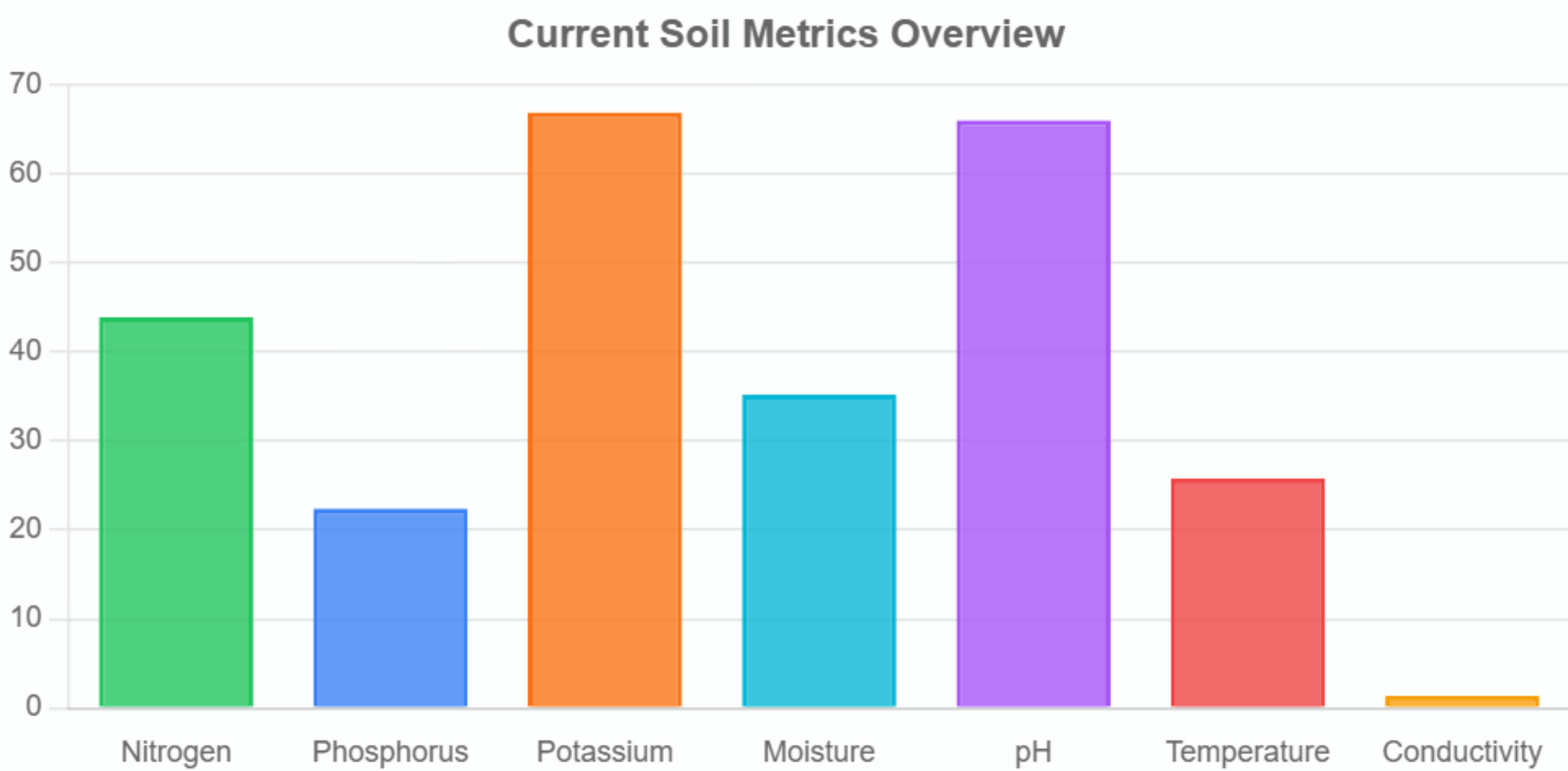


Results



Results

Soil Metrics Overview



Result 2:
The system dashboard shows real-time data like soil moisture, pH, temperature, and nutrients (NPK). The bar graph helps users quickly understand the soil condition.

Result 3:
The Arduino reads sensor data and controls the water pump using a relay. It turns the pump ON or OFF based on soil moisture levels, ensuring smart irrigation and water saving.

Conclusion

- The Smart Irrigation System worked well. It only watered the plants when needed, not all the time. This shows it used real sensor data to make smart decisions.
- Our idea was that the system would save water and improve how irrigation is done. The results proved this was true — water was not wasted, and the plants still got enough.
- This system is useful for farmers who want to save water, reduce costs, and grow crops in a smarter, more efficient way.

References

• **Arduino.** (n.d.). *Arduino Mega 2560 Rev3*. Retrieved from: <https://store.arduino.cc/products/arduino-mega-2560-rev3>

• **Gaurav, A., & Kaur, R.** (2020). *Smart Irrigation Using Soil Moisture Sensor*. International Journal of Scientific Research in Computer Science, Engineering and IT, 6(3), 100–105.

• **TinyGPS++ Library.** (n.d.). *GPS Library for Arduino*. Retrieved from: <https://github.com/mikalhart/TinyGPSPlus>

• **Smart Irrigation Website.** (2025). *Final project site*. Retrieved from: <https://friendly-granita-83cc22.netlify.app/>

Future Work

Adding agromarket: this will be to help farmers sell their produces on the same platform.

•**Adding agroanimal:** this is to help in tracking animals data for animal farmers with digital ear tags.

•**Train the users:** Teach farmers how to use the system so they can get the most out of it and farm in a more sustainable way.

•**Adding the system linking farm application suppliers and farmers:** here we are looking forward to see that farmers can buy farm supplies from the same web system hence helping and providing a one stop agricultural solution.